Project 1: Exploratory Data Analysis (EDA) and Visualization on Diabetes Dataset

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Objective

Practice data loading, visualization, and EDA using Python on the Pima Indians Diabetes dataset (diabetes.csv).

Part 1: Data Loading

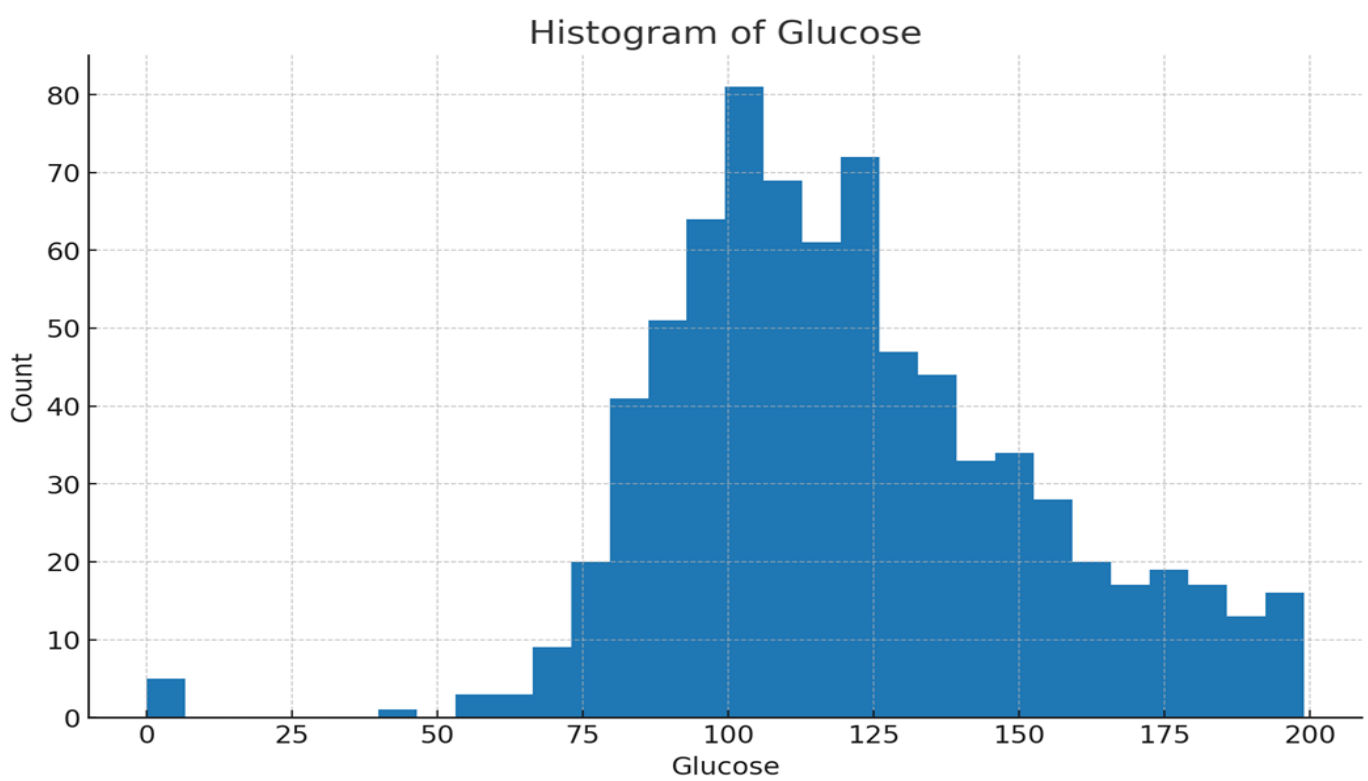
Shape: 768 rows × 9 columns.

Columns: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome

Part 2: Data Visualization

1) Histogram of Glucose

The distribution is right-skewed; most values are between ~80–150.

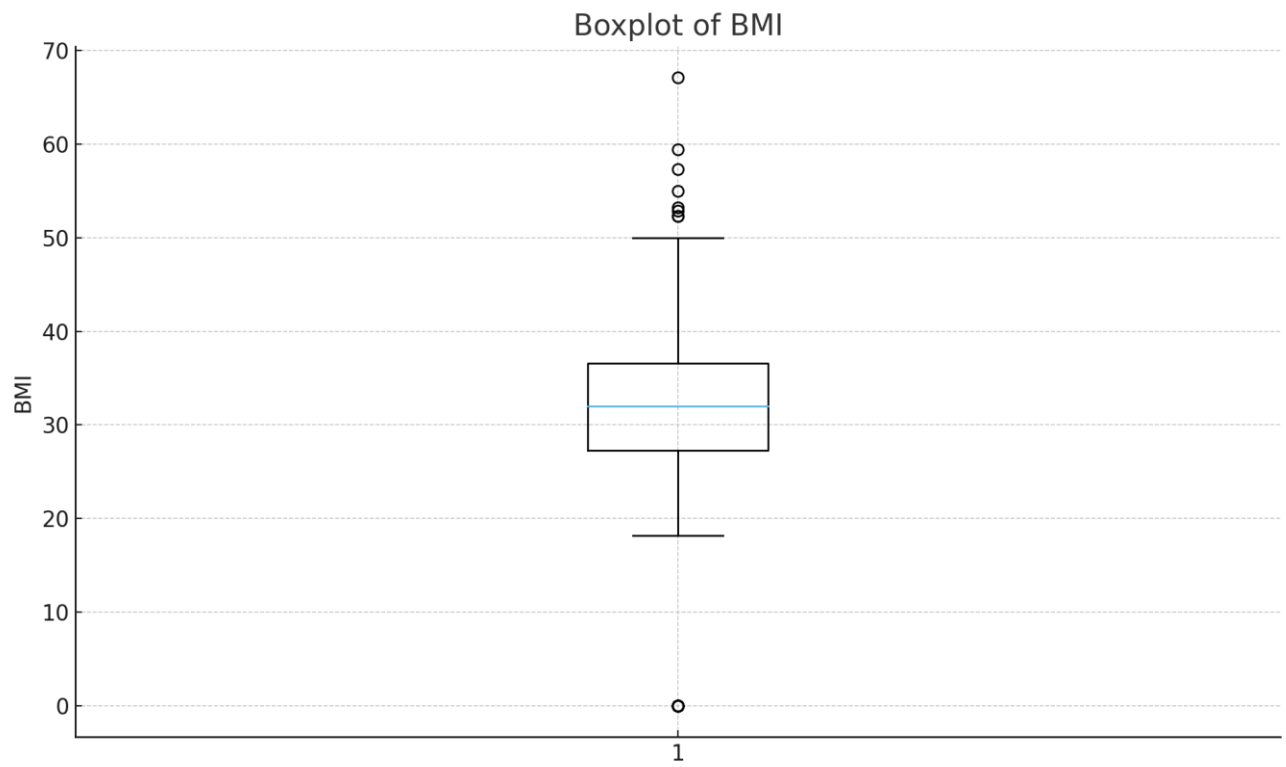


1. **Histogram of Glucose**

- Distribution is **right-skewed**.
- Most patients have glucose levels between 80–150.

2) Boxplot of BMI

Median BMI ≈ 30 with high-end outliers; indicates common obesity in the cohort.

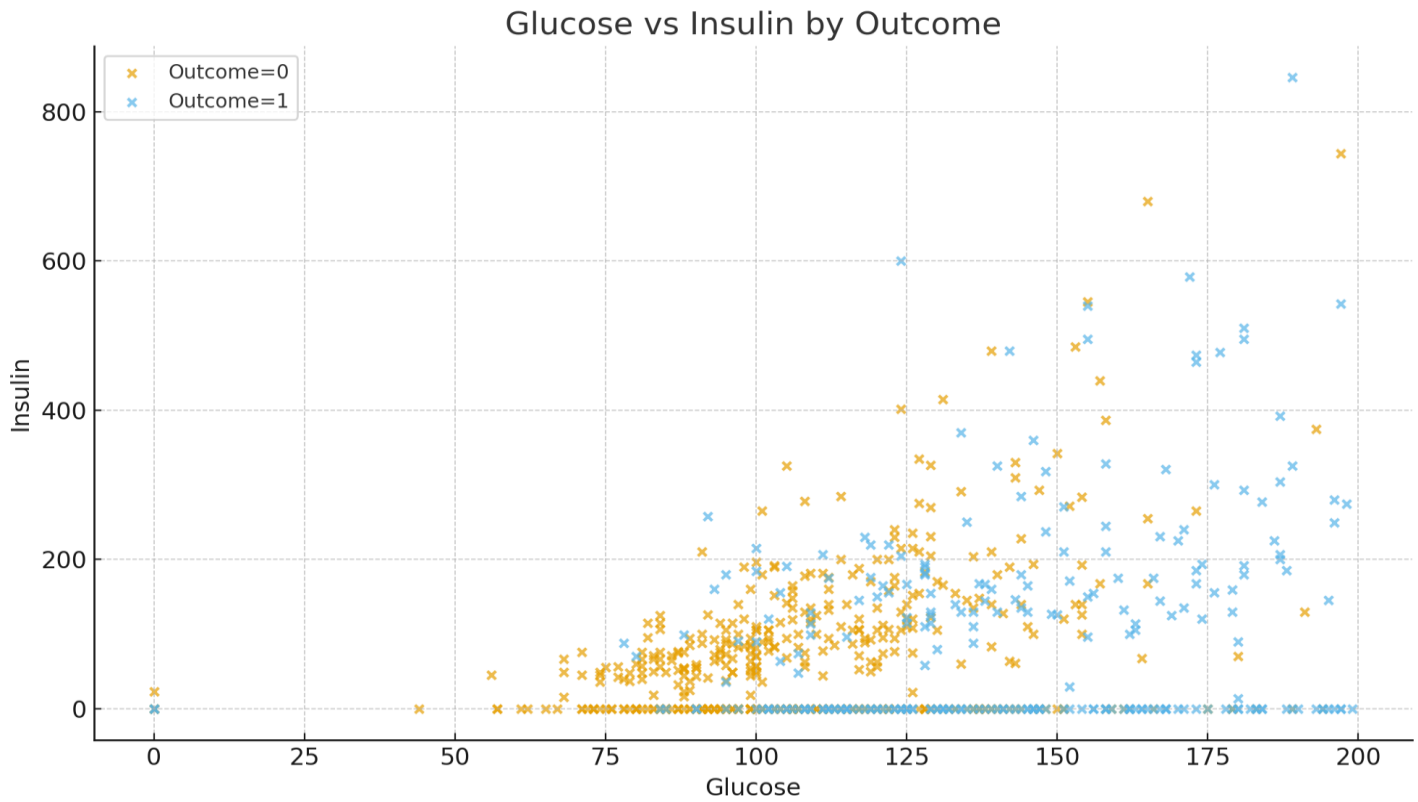


. Boxplot of BMI

- Median BMI ~30.
- Outliers exist at higher BMI values (>50).
- Suggests obesity are common in this dataset.

3) Scatter: Glucose vs Insulin by Outcome

Higher glucose tends to align with higher insulin; many zero insulin values likely indicate missing entries.

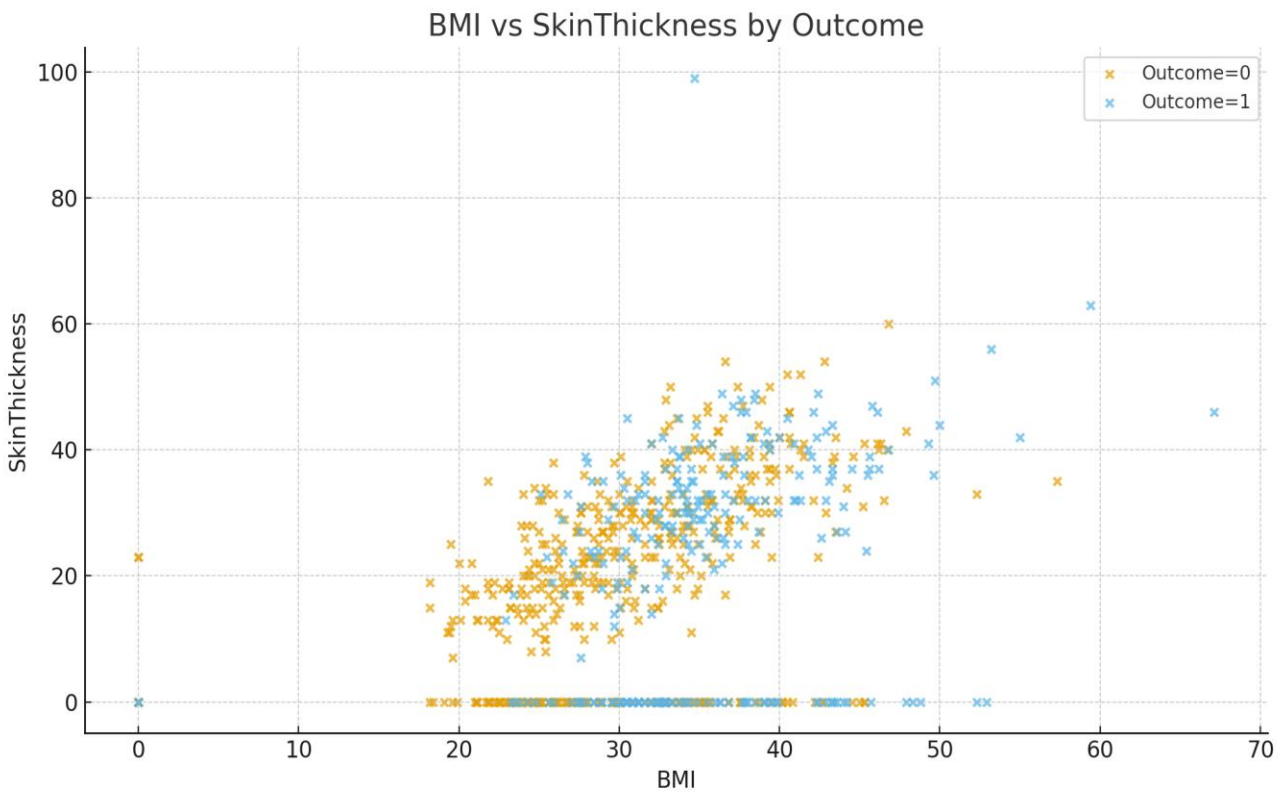


Scatter Plot: Glucose vs Insulin

- Positive relationship observed.
- Many entries have Insulin = 0 (likely missing/invalid values).
- Diabetic patients cluster at higher Glucose and Insulin.

4) Scatter: BMI vs SkinThickness by Outcome

A positive association is visible: higher BMI often corresponds to greater skinfold thickness.



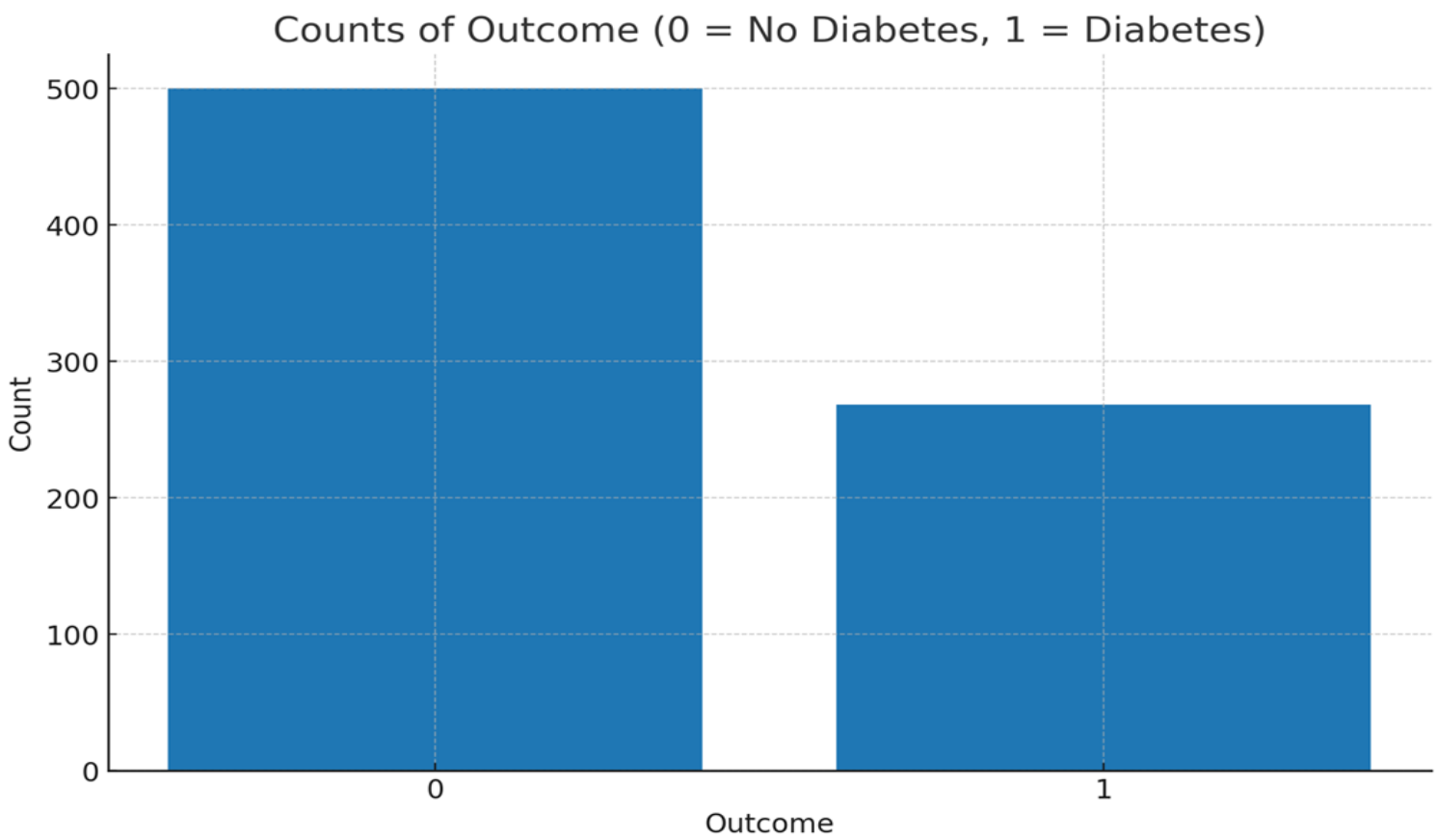
Relationship check: zeros in insulin indicate potential missing/unreliable entries.

4. Scatter Plot: BMI vs SkinThickness

- Positive trend between BMI and SkinThickness.
- Higher BMI patients often have thicker skinfold measurements.

5) Count Plot of Outcome

Non-diabetic patients (Outcome=0) are more frequent than diabetic patients (Outcome=1).



Part 3: Exploratory Data Analysis (EDA)

Summary statistics for numerical variables (excluding Outcome):

Feature	Mean	Median	Range	IQR	Variance	Std Dev
Pregnancies	3.85	3.00	17.00	5.00	11.35	3.37
Glucose	120.89	117.00	199.00	41.25	1022.25	31.97
BloodPressure	69.11	72.00	122.00	18.00	374.65	19.36
SkinThickness	20.54	23.00	99.00	32.00	254.47	15.95
Insulin	79.80	30.50	846.00	127.25	13281.18	115.24
BMI	31.99	32.00	67.10	9.30	62.16	7.88
DiabetesPedigreeFunction	0.47	0.37	2.34	0.38	0.11	0.33
Age	33.24	29.00	60.00	17.00	138.30	11.76

Full covariance and correlation matrices are included in the notebook.

1. Mean: Average value across all records. For example, average glucose is ~121, suggesting mild hyperglycemia in the sample.

2. Median: Middle value when data is sorted. For Glucose (117) vs mean (121), values are slightly skewed right.

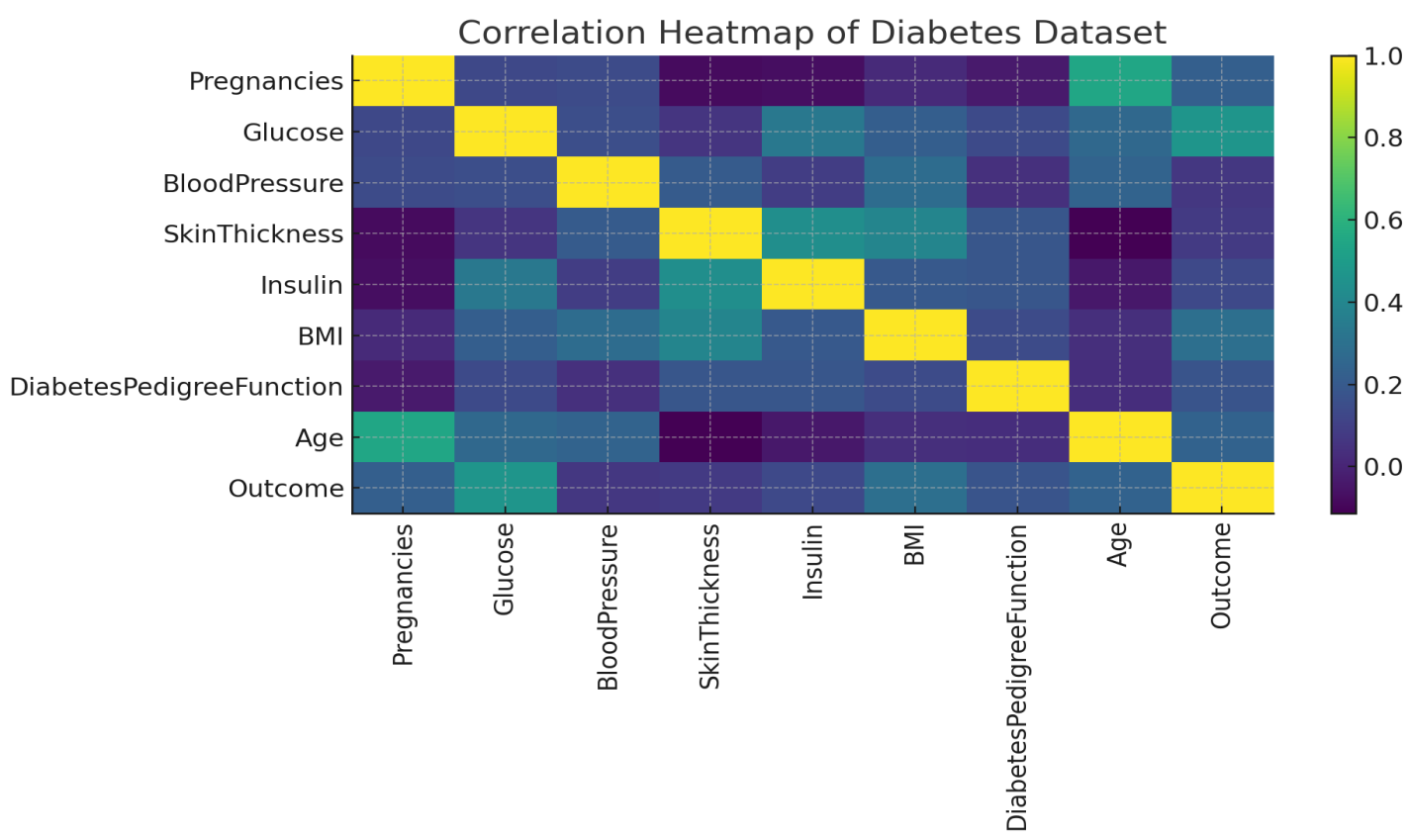
3. Range: Difference between max and min. Insulin has a huge range (846), reflecting extreme outliers and many zero entries.

4. IQR (Interquartile Range): Spread of the middle 50% of data. Insulin's IQR = 127.25, much larger than other variables, again showing wide variability.

5. Variance & Standard Deviation: Variability measures. Larger Standard Deviation (e.g., Insulin = 115.24) means the data is highly spread out. Smaller ones (e.g., DPF = 0.33) mean tighter clustering.

Part 4: a. Correlation Analysis

Glucose shows the strongest positive correlation with Outcome, followed by BMI and Age. BloodPressure and SkinThickness are weakly correlated with Outcome.



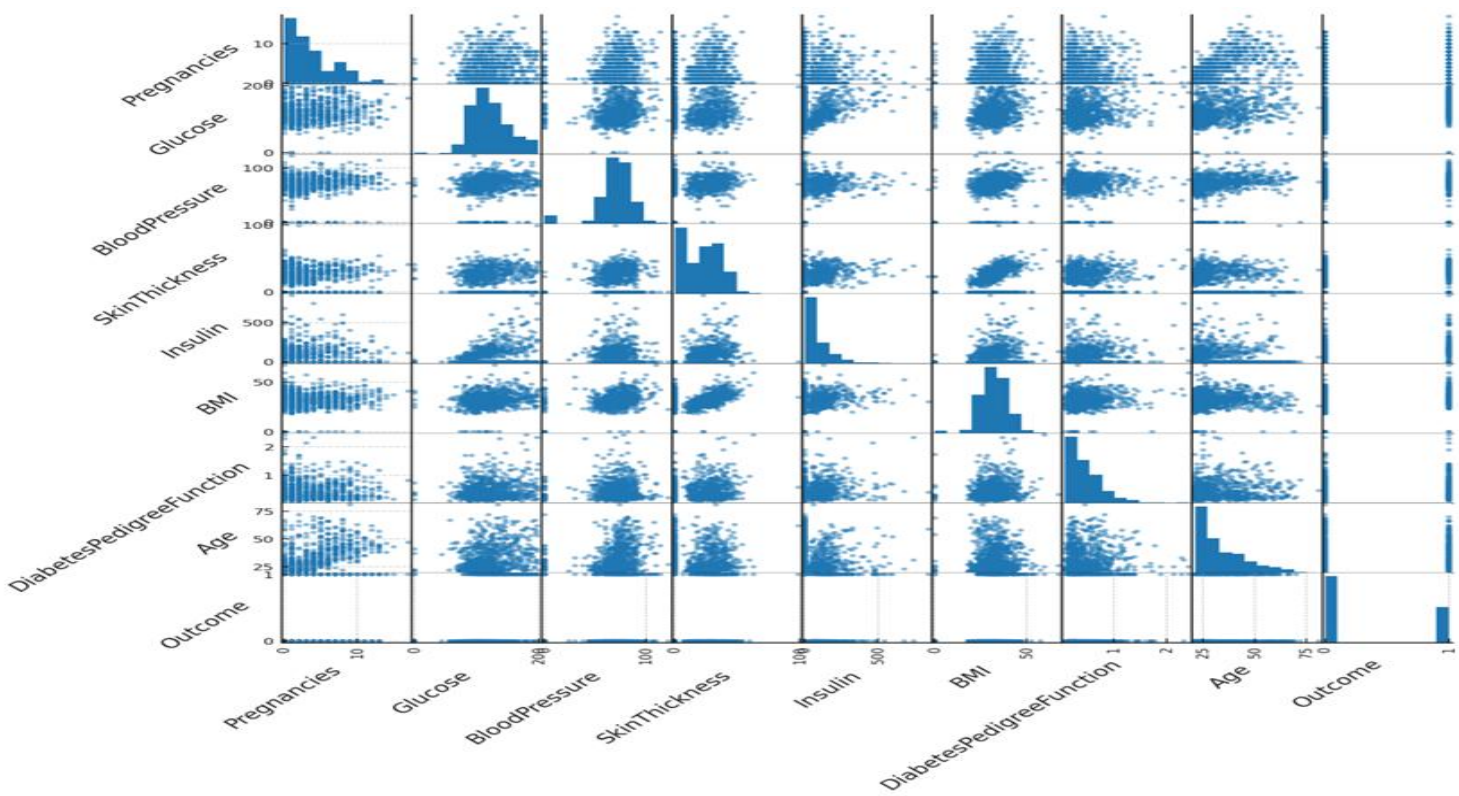
A full pairplot (multi-plot grid) is included in the Jupiter notebook.

Covariance Matrix: Shows how variables vary together (scale dependent).

Correlation Heatmap:

- Strong positive correlation: Glucose & Outcome (~0.47) → higher glucose strongly linked to diabetes.
- Moderate correlation: BMI & Outcome (~0.29), Age & Outcome (~0.24).
- Weak/near-zero correlation: BloodPressure, SkinThickness.

Part 4: b. Plot a pair plot of the dataset.

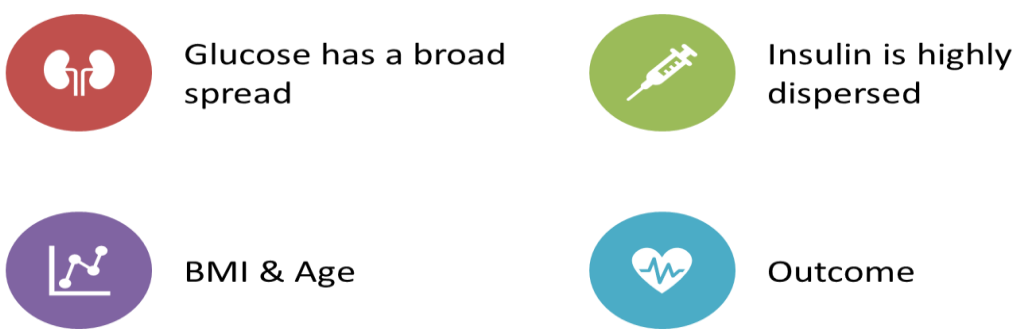


All pairs visualized at once; diagonals show distributions.

Clear separation in Glucose vs Outcome and BMI vs Outcome.

Overlaps in variables like BloodPressure, indicating weaker predictive power.

Observation



1. **Glucose** has a broad spread (Range = 199, Standard Deviation ~32) → aligns with diabetes being strongly tied to blood glucose variability.
2. **Insulin** is highly dispersed (Range = 846, Standard Deviation ~115) → confirms missing/zero issues; needs cleaning before modeling.
3. **BMI & Age** show moderate spread, useful predictors.
4. **Outcome** has mean ~0.35 → dataset is imbalanced (35% diabetic vs 65% non-diabetic).

Recommendations

1. Clean and preprocess insulin values (handle missing/zeros).
2. Prioritize Glucose, BMI, and Age for feature selection.
3. Use stratified sampling or class weighting to balance modeling.
4. Explore interactions (e.g., Age × BMI) and external factors.

Conclusion

Higher Glucose, BMI, and Age are associated with diabetes in this dataset. Insulin values show high variability with many zeros, suggesting potential missingness. BloodPressure and SkinThickness appear less predictive of Outcome.